

CAAS run, job 39361 - analysis

Run: 9 September 2026, student-caas-headnode, 8 MPI ranks across two compute nodes (student-caas-n01, student-caas-n02). $n = 130,000,000$. Output verified byte identical to the serial reference.

Serial baseline on the cluster: 78.29 s (77.60 compute + 0.69 write).

p	Total (s)	Speedup vs serial	Speedup vs p=1	Amdahl	Gap	kappa(p)	imbalance
1	52.50	1.49x	1.00x	1.00	0.0%	0.00015	1.000
2	26.86	2.91x	1.95x	1.96	0.4%	0.00072	1.001
4	13.99	5.60x	3.75x	3.78	0.7%	0.00116	1.001
8	7.59	10.31x	6.91x	7.05	1.9%	0.00124	1.001

Measured fractions: $r_p = 0.9819$, $r_s = 0.0179$, ceiling 55.9x.
Parallel efficiency at $p = 8$: 86.4%.

Hybrid, 2 processes x 2 threads, one process per node: 13.96 s, against 13.99 s for pure MPI at 4 processes. Indistinguishable at matched width, which is consistent with the laptop finding that arrangement only matters when the partition is unbalanced.

Three findings

1. Near linear scaling. 86.4% efficiency at 8 processes across two physical machines, and the curve had not flattened. The laptop plateaued near 7x because of the M3 Max's 4 efficiency cores and oversubscription past 14 workers. Cluster cores are homogeneous and dedicated, so that ceiling is gone.
2. Amdahl predicts to within 2%. On the laptop, theory overshoot measurement by roughly 23% at $p = 8$, and we attributed that to core heterogeneity rather than to the model. This run tests that claim: on homogeneous dedicated cores at a balanced process count, the same model gives 7.05 against a measured 6.91. The gap was hardware, and we can now show it rather than argue it.
3. Our prediction about communication was wrong. We said a two node run was the missing measurement and that kappa would finally dominate. The gather did get genuinely dearer, 0.0633 s at $p = 8$ against 0.0138 s at the same width on one machine, 4.6x larger, which confirms it is real network traffic now. But as a fraction of runtime kappa is still only 0.0012.

The reason is a ratio we had not considered carefully: about 59 MB of primes moved once, against 52 seconds of arithmetic, at roughly 0.9 GB/s effective throughput. Communication would only dominate at a much smaller n , with far more ranks, or on a slower interconnect.

What this run does not show

Process counts were 1, 2, 4 and 8, all powers of two, chosen so the partition would be balanced and the scaling result clean. Every imbalance ratio therefore came back at 1.00, and the residue-class saw tooth does not appear in this data. That finding still rests on the laptop sweep across 18 widths. Re-running at $p = 3, 6$ or 12 would confirm it here, and should, since it comes from arithmetic in the partitioning rather than from hardware.

Raw output: caas_results.txt.